ABRAHAM TOLUWASE OWODUNNI

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RESEARCH INTERESTS

Token-Efficient Language Models, LLM Efficiency, Tokenization, Representation Learning

EDUCATION

Ph.D. in Computer Science and Engineering, The Ohio State University 2024 – Present
Advisor: Prof. Sachin Kumar

B.Sc. in Biochemistry, University of Ilorin 2015 – 2019

SELECTED PUBLICATIONS

* denotes equal contribution

- **Owodunni, A.T.**, . . . , Limisiewicz, T., and Kumar, S. Dynamic Multi-Byte Prediction With Hierarchical Language Models (2026). ***Under Review, NAACL***.
- **Owodunni, A.T.**, Ahia, O., and Kumar, S. FlexiTokens: Flexible Tokenization for Evolving Language Models (2026). ***ACL***.
- **Owodunni, A.T.** and Kumar, S. Continually Adding New Languages to LLM with Vector Steering (2026). ***TMLR***.
- **Owodunni, A.T.***, et al. AfriMed-QA: A Pan-African, Multi-Specialty, Medical Question-Answering Benchmark Dataset (2025). ***ACL, Best Social Impact Award***.
- **Owodunni, A.T.***, et al. AccentFold: A Linguistic Journey for sample-efficient Zero-Shot ASR Adaptation to Target Accents (2024). ***EACL***.
- Ogun, S., **Owodunni, A.T.**, et al. 1000 African Voices: Advancing Inclusive Multi-Speaker Multi-Accent Speech Synthesis with learning new speaker vectors (2024). ***Interspeech***.
- Afonja, A., . . . , **Owodunni, A.T.**, et al. Performant ASR Models for Medical Entities in Accented Speech (2024). ***Interspeech***.
- **Owodunni, A.T.** and Emezue, C.C. Koya: A Recommender System for Large Language Model Selection (2023). ***AfricaNLP***.
- Oladipo, A., . . . , **Owodunni, A.T.**, Adelani, D.I., and Lin, J. Better Quality Pre-training Data and T5 Models for African Languages (2023). ***EMNLP***.
- Leong, C., . . . , **Owodunni, A.T.**, et al. Bloom Library: Multimodal Datasets in 300+ Languages for a Variety of Downstream Tasks (2022). ***EMNLP***.

PROJECTS

Recursive Latent Compression for Unbounded Long-Form Generation (Ongoing)

- Developing a learned **boundary predictor** that recursively compresses past context into **latent tokens**, letting a pretrained dense LLM generate beyond its context window
- Keeps generation within the model's trained **RoPE** range by compacting the oldest tokens 2× at the 64K limit, as a lightweight module added to an existing model

AfriMedVQA: African LLM Medical Benchmark (Ongoing)
Funded by Google and PATH ($500K)

- Co-leading creation of the first pan-African, multi-specialty, multimodal, multilingual medical QA benchmark

- Work featured in the **Google Health AI 2025** report

Tokenization Alignment for Stable Multi-Turn RL

- Identified **within-sequence tokenization drift** as a structurally distinct source of instability in multi-turn RL training
- Developed a context-aware preprocessing method aligning user tokens to adjacent assistant splits, outperforming standard incremental tokenization on a multi-turn GSM8K benchmark during Qwen3 **GRPO** training with no inference-time cost

Dynamic Tokenization for Images (Ongoing)

- Developing a **gradient-based, content-adaptive tokenization** method for images, replacing the fixed-size patch grids (e.g., 16×16) used in standard vision transformers

EXPERIENCE

Machine Learning Researcher | Intron Mar. 2023 – Aug. 2024

- **Led pioneer ASR team**, improving **medical WER** on out-of-distribution (OOD) African accents through accent clustering analysis and informed data sampling
- Improved ASR robustness by **15% medical WER** across OOD accents via multi-task learning on NER, training-data domain, and voice activity detection; deployed models to **20+** hospitals
- **Co-invented a patented** data augmentation technique for low-resource speech processing
- Designed and delivered masterclasses training **70+ practitioners** on building speech technologies

Researcher | Masakhane May 2021 – Present

- Built models for **360+ languages** spanning visual storytelling, language modeling, and ASR
- Designed a recommender system using token masking to score model–language compatibility and predict downstream task performance
- Moderate community talks, sessions, and webinars on NLP for under-resourced languages

Data Analyst | FHI 360 Jan. 2020 – Dec. 2020

- Built end-to-end analytics workflows transforming raw **health data** into stakeholder-facing reports
- Administered the national DHIS 2.0 health database with a 5-person team
- Automated recurring reporting pipelines, reducing analysis time by over 20%

SERVICE & OUTREACH

- **National AI Policy Contributor**, NITDA Nigeria: volunteer expert participating in the design of Nigeria's National Artificial Intelligence Policy
- **Publication Chair**: MRL Workshop @ EMNLP 2024; MLGH Workshop @ ICLR 2023
- **Lead Organizer**: MLC-Africa (50+ mentees; 300+ community members); Data Science for Health Workshop @ Deep Learning Indaba: 2022, 2023
- **Reviewer**: ICLR 2025, ACL 2026, EMNLP 2026, AfricaNLP 2024, 2025

TALKS & TEACHING

- **TA:** CSE 5525: Foundations of Speech and Language Processing
- **CSE 5525 Guest Lecture:** How to train Llama 5 with 500B parameters
- **PyOhio 2026**: LLM PreTokenization: Free Gains You Leave on the Table
- **Invited Talk**: "Going Beyond Text with AI for African Languages", Center for African Studies